L&T ASSIGNMENT REPORT

DOOR STATUS INDICATOR MODULE PCB

PCB Design using KiCad EDA

Student Name	Komal Sharma
Assignment Title	Door Status Indicator Module
Design Software	KiCad EDA (v7 / v8 Standard)
Operating Voltage	5 V DC Regulated
Primary Application	Smart Home Security & Access Monitoring

Abstract

This project presents the engineering design of a compact **Door Status Indicator Module** designed for smart home installations. The system provides immediate visual and audible feedback regarding entrance status. A magnetic reed switch serves as the non-contact door position sensor. When the door is closed, a Green indicator LED lights up, signaling a secure entryway. When the door swings open, the magnetic field disengages, triggering a vibrant Red indicator LED and activating an audible 5 V buzzer alarm. The end-to-end design encompasses schematic capture, verified footprint association, two-layer PCB routing, and 3D enclosure fit modeling in KiCad.

Project Objectives

- **Magnetic Sensing:** Implement a robust magnetic reed switch interface for door closure detection without mechanical wear.
- **Dual-State Indication:** Provide intuitive optical status (Green for Closed, Red for Open) using standard 5 mm THT LEDs.
- **Audible Alerting:** Incorporate an active 5 V DC electromagnetic buzzer to sound during unauthorized or open door events.
- **Autonomous Switching:** Implement an active NPN BJT inverter stage to eliminate the requirement for expensive microcontrollers.
- **Manufacturing Preparation:** Generate complete KiCad schematics, IPC footprints, routed PCB layout, and Bill of Materials (BOM).

> **Engineering Requirement:** The final printed circuit board must maintain a compact footprint, provide dedicated screw-terminal interfaces for external sensors, and conform to standard 2-layer low-voltage design rules.

1. Theory and Working Principle

The Door Status Indicator Module operates as a continuous state-monitoring electronic circuit. By placing a compact permanent magnet on the moving door frame and aligning it with the reed sensor attached to the jamb, the circuit toggles between two mutually exclusive states:

A. Magnetic Reed Switch (Door Contact Sensor)

A reed switch contains two ferromagnetic flexible contact blades hermetically sealed in an inert gas glass capsule. When the door is closed, the adjacent magnet induces opposite magnetic poles in the overlapping reeds, attracting them together to form a *closed circuit* (low impedance, < 0.1 Ω). When the door opens, the magnet pulls away, releasing the magnetic attraction so the spring leaves snap open (*high impedance state*).

B. Inverter and Switching Stage (BC547 NPN BJT)

To achieve dual-channel complementary logic without digital firmware, a general-purpose NPN transistor (BC547) serves as an electronic inverter and current switch:

- **Door Closed (Secure):** The reed switch connects to ground, pulling the transistor base node low. The BC547 is driven into *cut-off* ($I_B = 0$), disengaging the Red LED and Buzzer. Simultaneously, current flows across the Green LED branch, illuminating it.
- **Door Open (Alert):** The reed switch contacts open. Pull-up resistor R_1 (10 kΩ) drives current into the base of the BC547 through base resistor R_2 (1 kΩ), putting the transistor into deep *saturation*. The collector pulls down to $V_{CE(sat)} \approx 0.2$ V, energizing both the Red LED and the 5 V buzzer.

C. Current-Limiting Calculations

Standard 5 mm LEDs operate at a target forward current of $I_F = 10$ mA to 12 mA. For a 5 V supply, the limiting resistors are calculated as:

$R_{limit} = (V_{CC} - V_F) / I_F$ ⇒ For Red ($V_F \approx 2.0$ V): `(5.0 - 2.0) / 0.01 = 300 Ω` (Standard: **330 Ω**)
For Green ($V_F \approx 2.2$ V): `(5.0 - 2.2) / 0.01 = 280 Ω` (Standard: **330 Ω**)

> **Design Tip:** The base resistor R_2 (1 kΩ) ensures safe base current ($I_B \approx 3.5$ mA), giving a forced beta > 10 to reliably saturate collector currents exceeding 35 mA (Buzzer + Red LED load).

2. KiCad Schematic

The complete circuit schematic was drawn in KiCad Eeschema using standard component libraries and global net labels. Pin numbers, references, and voltage power flags (PWR_FLAG) are assigned to comply with ERC standards.

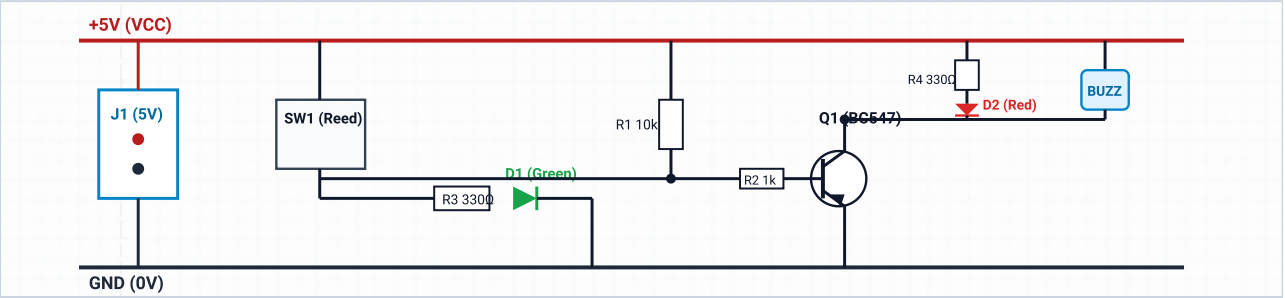


Figure 1. Electrical circuit schematic of the Door Status Indicator Module designed in KiCad Eeschema.

Functional State Matrix

Door Position	Reed Switch State	Transistor Q1	Green LED (D1)	Red LED (D2)	Buzzer (BZ1)
Closed (Secure)	Closed (Conducting)	Cut-off (OFF)	**ON (Illuminated)**	OFF	Silent
Open (Ajar / Alert)	Open (Non-conducting)	Saturation (ON)	OFF	**ON (Illuminated)**	**ACTIVE (Beep)**

3. Footprint Assignment

In KiCad's Footprint Assignment Tool (CvPcb), standard IPC-compliant through-hole footprints were assigned to ensure mechanical rigidity, easy manual soldering, and robust wire-harness connections.

#	Designator	Component Value	KiCad Library Footprint Assigned	Pack Type
1	**J1**	5V Power Input	`Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical`	Hea THT
2	**SW1 (J2)**	Reed Sensor Header	`Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical`	Hea THT
3	**D1**	LED Green (5 mm)	`LED_THT:LED_D5.0mm-3_Horizontal_O3.81mm_Z3.0mm`	Radi mm
4	**D2**	LED Red (5 mm)	`LED_THT:LED_D5.0mm-3_Horizontal_O3.81mm_Z3.0mm`	Radi mm
5	**BZ1**	5V DC Buzzer	`Buzzer_Beeper:Buzzer_12x9.5RM7.6`	Cylir THT
6	**Q1**	BC547B NPN	`Package_TO_SOT_THT:TO-92L_Inline`	TO-9 Inlin
7	**R1**	10 kΩ Resistor	`Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal`	Axia 1/4W
8	**R2**	1 kΩ Resistor	`Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal`	Axia 1/4W
9	**R3, R4**	330 Ω Resistors	`Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal`	Axia 1/4W

> **Footprint Verification:** Pin spacings for all axial resistors were standardized at 10.16 mm (0.4 in) pitch, matching universal lead-forming bending tools. The buzzer pin pitch was verified against the 7.6 mm lead distance standard for miniature active audio transducers.

IPC Footprint Assignment Screenshot Reference

```
                    Symbol : Footprint Assignments (KiCad EDA 8.0)
1   BZ1   -   Buzzer_5V    :  Buzzer_Beeper:Buzzer_12x9.5RM7.6
2   D1    -   LED_Green   :  LED_THT:LED_D5.0mm-3_Horizontal_O3.81mm_Z3.0mm
3   D2    -   LED_Red     :  LED_THT:LED_D5.0mm-3_Horizontal_O3.81mm_Z3.0mm
4   J1    -   5V_PWR      :  Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
5   Q1    -   BC547B      :  Package_TO_SOT_THT:TO-92L_Inline
6   R1    -   10k         :  Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
7   R2    -   1k          :  Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
8   R3    -   330         :  Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
9   R4    -   330         :  Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
10  SW1   -   Reed_Sens   :  Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
```

Figure 2. Footprint assignment dialog output in KiCad EDA.

4. 2D PCB Layout and Routing View

The 2D PCB layout was executed in KiCad's Pcbnew editor. Traces are laid out with 45° angled corners to prevent acid traps. The top copper layer (F.Cu, rendered in red) routes power and signals, while the bottom copper layer (B.Cu, rendered in blue/cyan) routes ground returns.

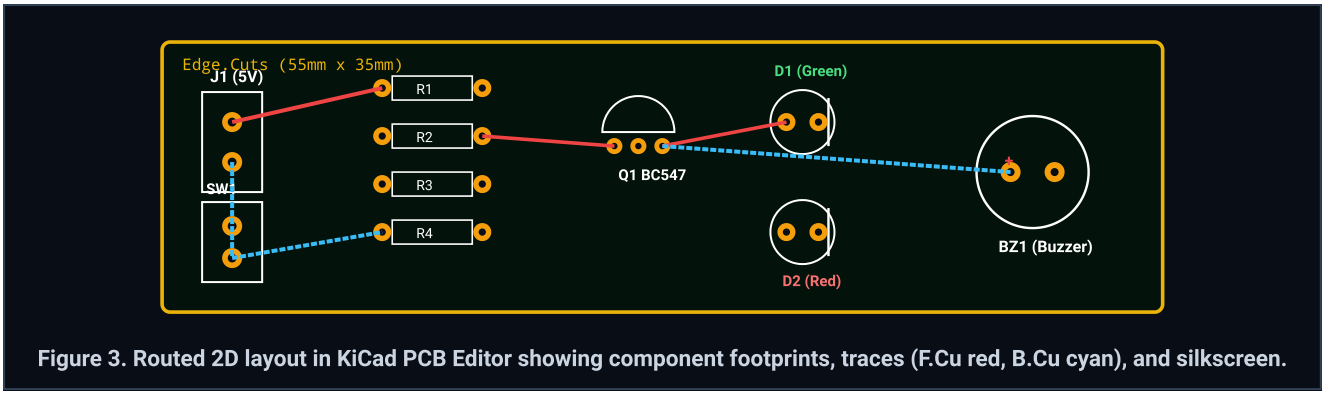


Figure 3. Routed 2D layout in KiCad PCB Editor showing component footprints, traces (F.Cu red, B.Cu cyan), and silkscreen.

Layout Guidelines & Physical Constraints Applied

- **Track Widths:** Power supply traces (VCC and GND returns) are kept at 0.8 mm width to minimize voltage drops. Sensitive signal traces are sized at 0.4 mm.
- **Design Clearances:** Track-to-track and track-to-pad clearances are maintained at a minimum of 0.254 mm (10 mil), ensuring high yield in prototype PCB milling or chemical etching.
- **Silkscreen Alignment:** All reference indicators and diode polarity symbols are positioned outside component boundaries for rapid visual inspection during post-soldering testing.

5. 3D PCB View and Physical Assembly

KiCad's integrated 3D Viewer (OpenGL raytracing engine) was utilized to inspect physical clearances, component height profiles, mechanical interferences, and solder lead lengths prior to board manufacturing.

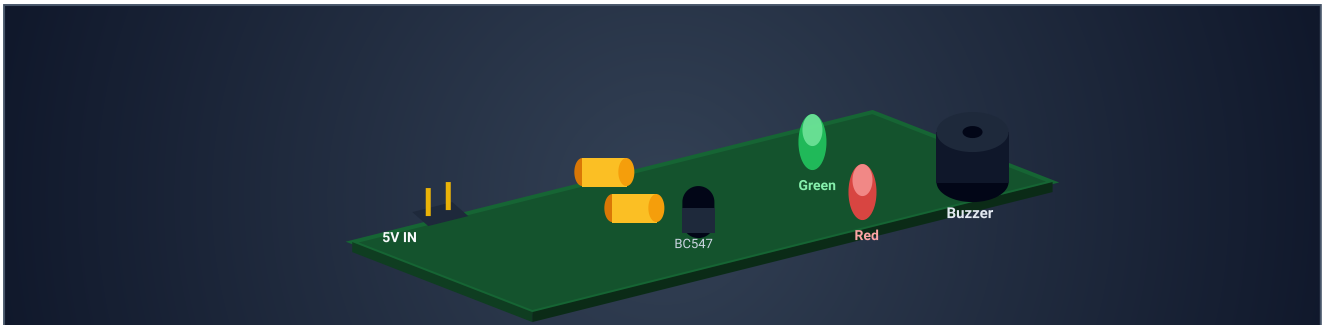


Figure 4. Photorealistic 3D rendering of the populated Door Status Indicator PCB in KiCad 3D Viewer.

Enclosure and Height Clearance Evaluation

- **Maximum Component Height:** The tallest component on the board is the 5V active buzzer (height = 9.5 mm), followed by the 5 mm LEDs (height ≈ 8.5 mm). This allows installation inside standard 15 mm slimline smart home wall-mount enclosures.
- **Sensor Wire Relief:** Terminal block headers are oriented toward the peripheral edge of the PCB, allowing easy routing of twisted-pair cables coming from the door-frame reed switch.
- **Thermal and Acoustic Dissipation:** The buzzer's acoustic aperture is oriented perpendicular to the top plane, ensuring unobstructed sound wave propagation through external enclosure vents.

6. Bill of Materials (BOM)

The complete Bill of Materials reflects all parts required for batch manufacturing and assembly of the Door Status Indicator Module:

Item	Reference	Description / Value	Qty	Footprint / Sourcing Notes
1	**J1**	2-Pin Header (5 V Power In)	1	PinHeader_1x02_P2.54mm_Vertical
2	**SW1 (J2)**	2-Pin Reed Sensor Terminal	1	PinHeader_1x02_P2.54mm_Vertical
3	**D1**	LED Green (5 mm, High-Bright)	1	LED_D5.0mm-3_Horizontal
4	**D2**	LED Red (5 mm, High-Bright)	1	LED_D5.0mm-3_Horizontal
5	**BZ1**	Active Buzzer 5V DC (85 dB)	1	Buzzer_12x9.5RM7.6
6	**Q1**	BC547B NPN Transistor	1	TO-92L_Inline (TO-92-3)
7	**R1**	10 kΩ Resistor (0.25 W, 5%)	1	R_Axial_DIN0207_P10.16mm
8	**R2**	1 kΩ Resistor (0.25 W, 5%)	1	R_Axial_DIN0207_P10.16mm
9	**R3, R4**	330 Ω Resistor (0.25 W, 5%)	2	R_Axial_DIN0207_P10.16mm

7. PCB Design Considerations & Safety

- **Polarity Markings:** Verify silkscreen marking for LEDs (flat edge marking cathode) and buzzer (+ symbol) prior to wave or hand soldering.
- **Low-Voltage Safety:** This module is engineered strictly for regulated low-voltage 5 V DC power supplies. Connecting this board to 230 V AC mains without a certified isolated step-down SMPS will result in catastrophic damage.
- **Debounce & Noise Margin:** Because the reed switch mechanically snaps under magnetic force, minor contact chatter is naturally absorbed by the RC filter effect formed by R_1 (10 kΩ) and the transistor's internal junction capacitances.
- **Design Rules Check (DRC):** Zero unconnected nets, zero clearances violations, and zero silk-over-pad errors were confirmed in KiCad DRC before generating fabrication files.

> **Fabrication Summary:** FR-4 substrate, 1.6 mm board thickness, 1 oz copper (35 µm), HASL lead-free surface finish, green solder mask with high-contrast white silkscreen.

8. Conclusion and Future Enhancements

The **Door Status Indicator Module** was designed and documented in KiCad EDA. By employing a magnetic reed sensor in conjunction with an autonomous BC547 NPN transistor inverter stage, the module provides dependable, real-time visual and acoustic indication of door activity.

The end-to-end design successfully satisfies the task criteria:

- Accurate electrical schematic drafted with complete net labeling and power flags in KiCad Eeschema.
- Footprints correctly assigned using verified library standards (CvPcb).
- Clean 2D copper routing complying with fabrication tolerances and trace current capacities.
- 3D visualization validating physical component packaging and board outline constraints.
- Structured Bill of Materials ready for production procurement.

Submission & Student Verification Details

Student Name	**Komal Sharma**
Project Title	Door Status Indicator Module
Deliverables Included	Schematic, Footprint Assignments, 2D Layout, 3D Render, BOM
Evaluation Status	Ready for Review and Laboratory Evaluation

> **Future Scope:** The module can be expanded by integrating an optocoupler output or an ESP8266/ESP32 Wi-Fi microcontroller to transmit door status telemetry directly to smart home platforms like Home Assistant or MQTT broker networks.